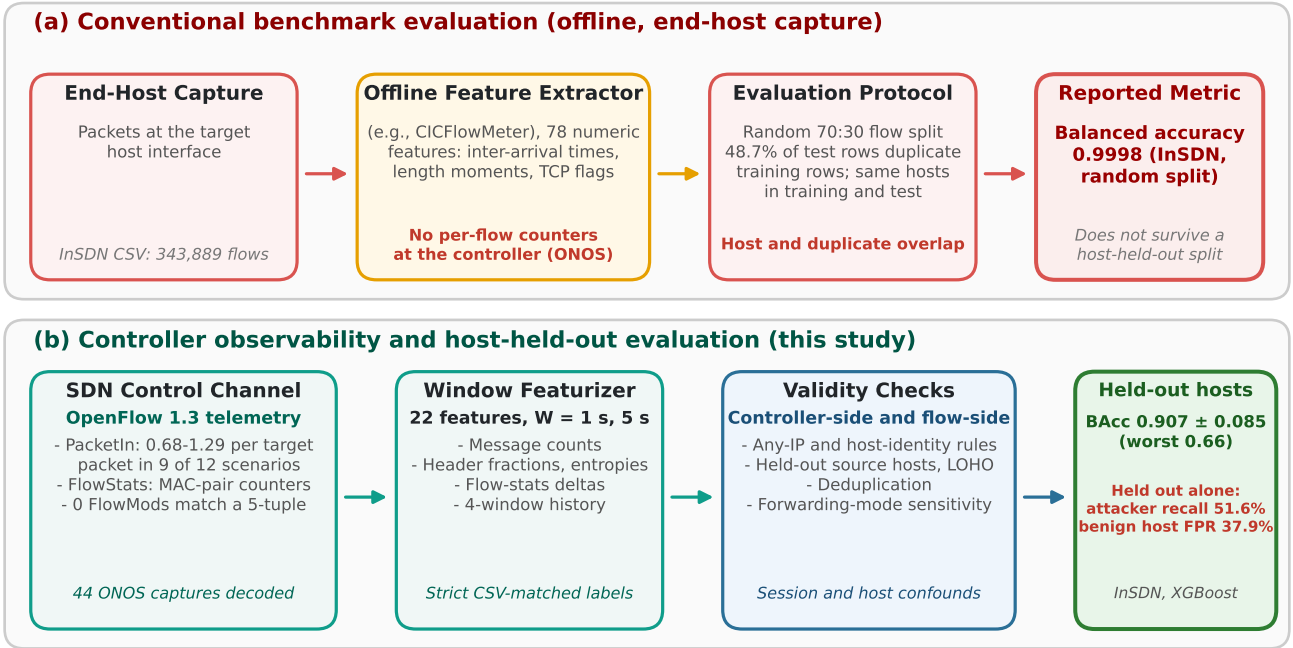

(a) Conventional benchmark evaluation (offline, end-host capture)
End-Host Capture
Packets at the target host interface
InSDN CSV: 343,889 flows
Offline Feature Extractor
(e.g., CICFlowMeter), 78 numeric features: inter-arrival times, length moments, TCP flags
No per-flow counters at the controller (ONOS)
Evaluation Protocol
Random 70:30 flow split
48.7% of test rows duplicate training rows; same hosts in training and test
Host and duplicate overlap
Reported Metric
Balanced accuracy 0.9998 (InSDN, random split)
Does not survive a host-held-out split
(b) Controller observability and host-held-out evaluation (this study)
SDN Control Channel
OpenFlow 1.3 telemetry
- PacketIn: 0.68-1.29 per target packet in 9 of 12 scenarios
- FlowStats: MAC-pair counters
- 0 FlowMods match a 5-tuple
44 ONOS captures decoded
Window Featurizer
22 features, W = 1 s, 5 s
- Message counts
- Header fractions, entropies
- Flow-stats deltas
- 4-window history
Strict CSV-matched labels
Validity Checks
Controller-side and flow-side
- Any-IP and host-identity rules
- Held-out source hosts, LOHO
- Deduplication
- Forwarding-mode sensitivity
Session and host confounds
Held-out hosts
BAcc 0.907 ± 0.085 (worst 0.66)
Held out alone: attacker recall 51.6% benign host FPR 37.9%
InSDN, XGBoost